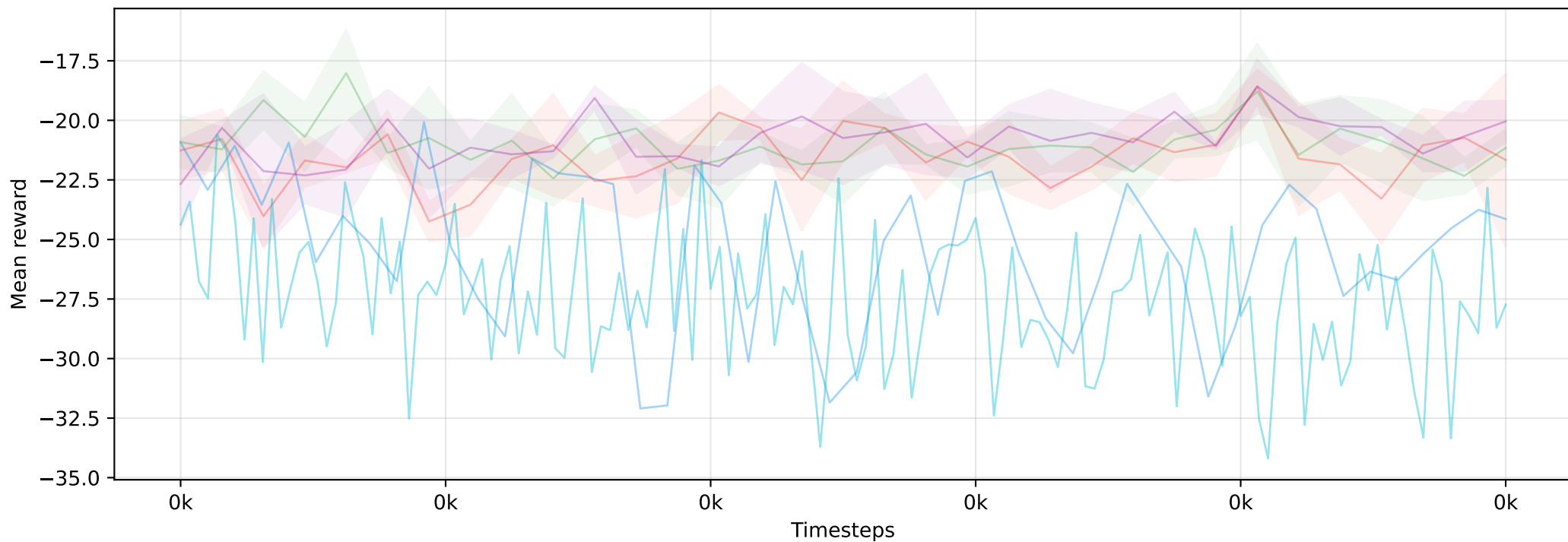
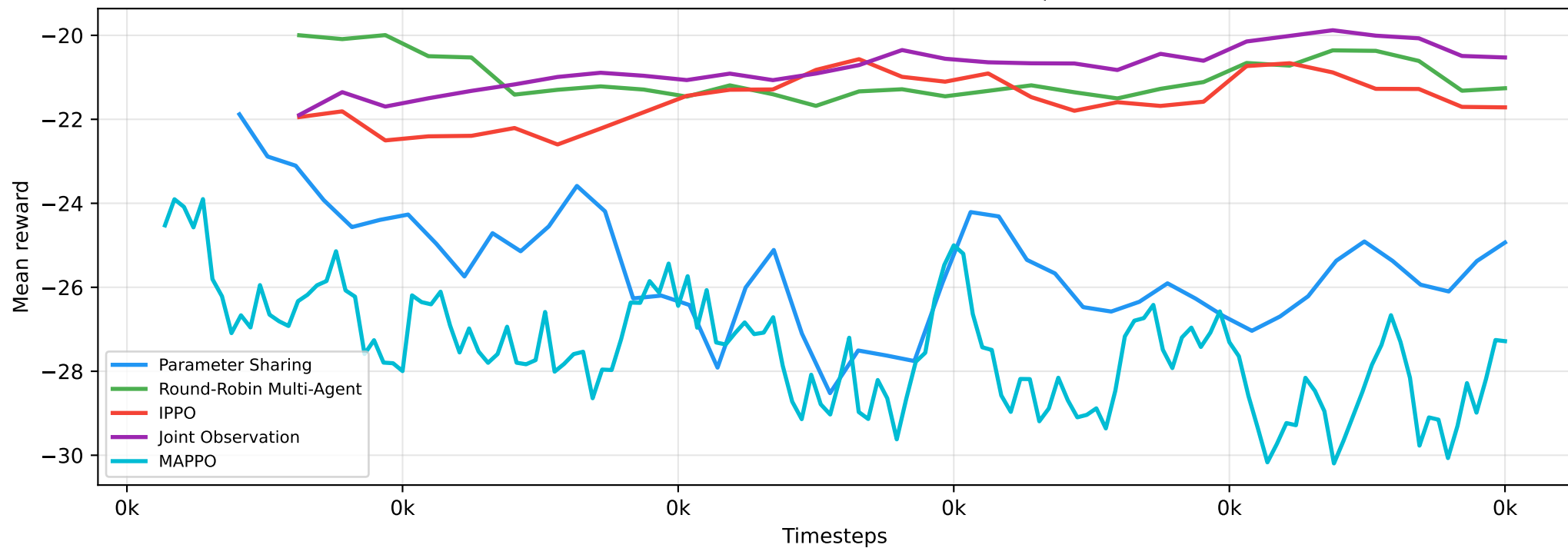


# PPO Evaluation — Mean Episode Reward

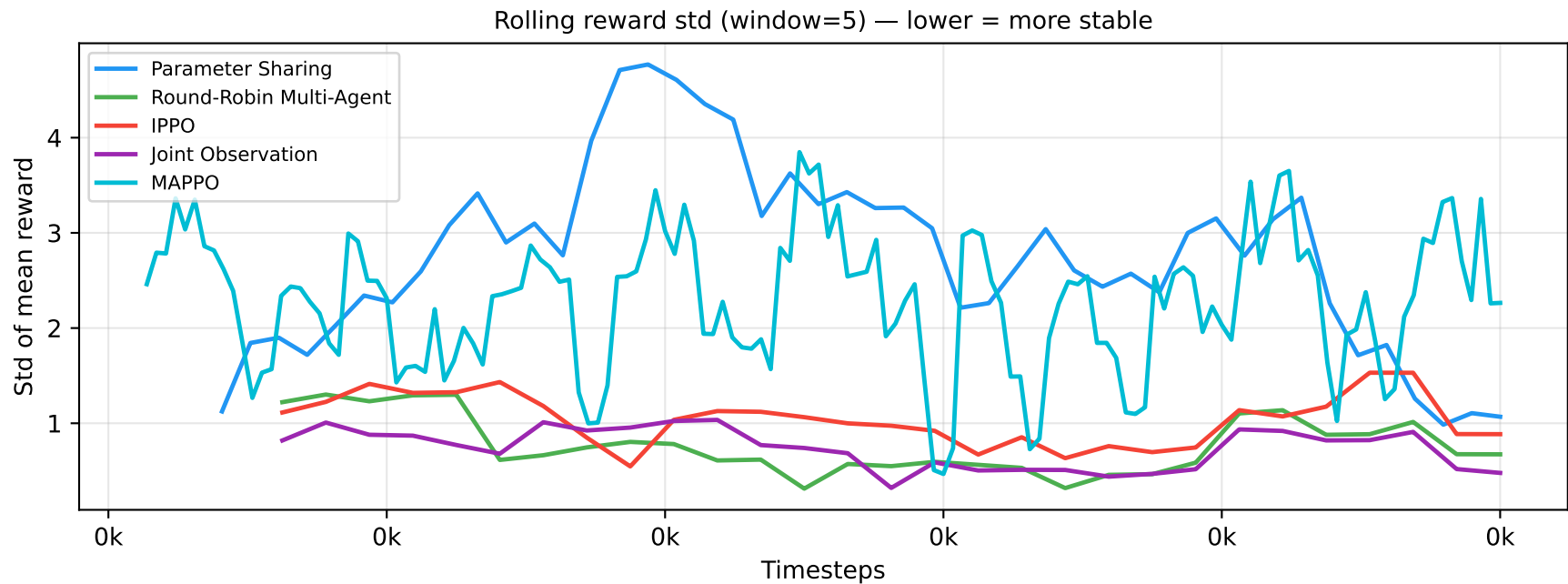
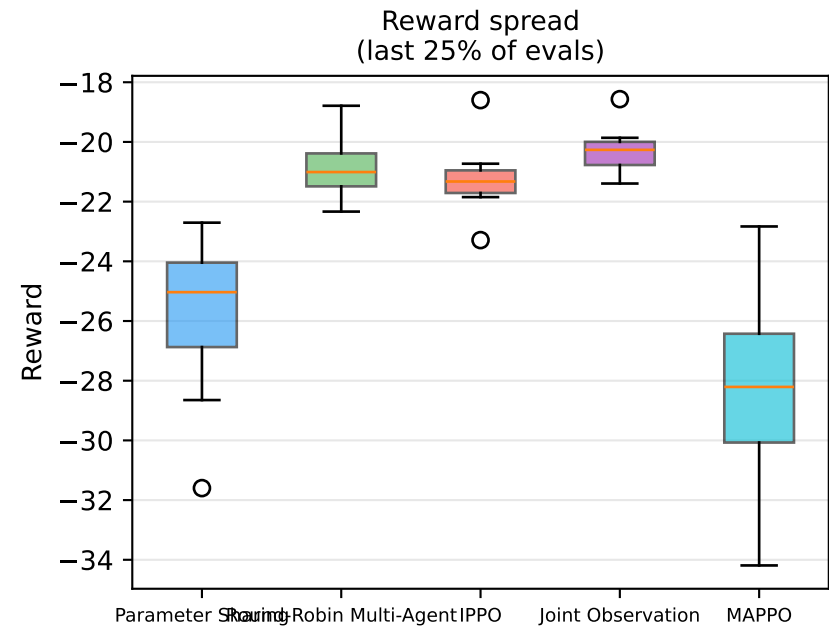
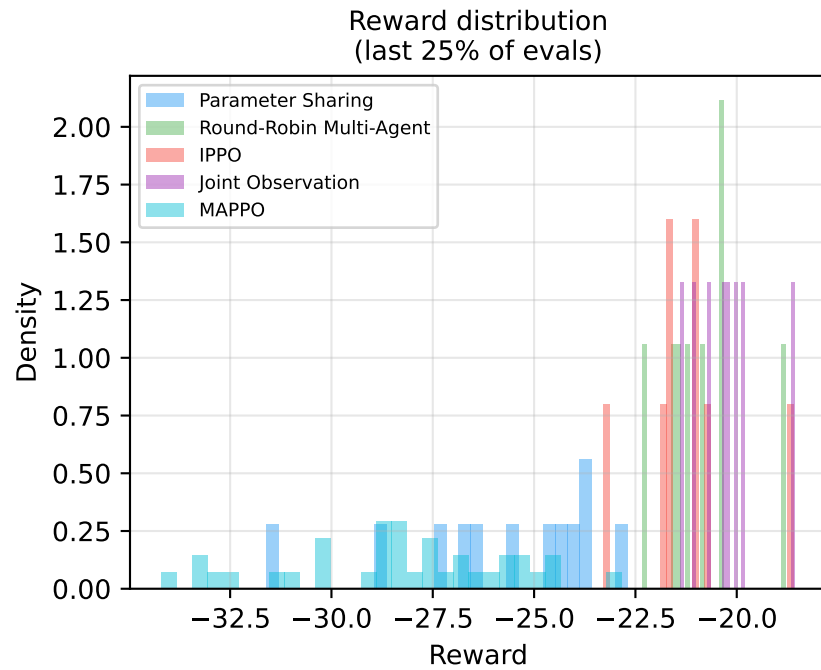
Raw eval mean reward ( $\pm 1$  std)



Smoothed eval mean reward (window=5 checkpoints)



# PPO Evaluation — Distribution & Convergence



## PPO Evaluation — Summary Statistics (last 25% of evals)

| Run                   | Total steps | Eval checkpoints | Mean reward | Std   | Max     | Min     |
|-----------------------|-------------|------------------|-------------|-------|---------|---------|
| Parameter Sharing     | 0k          | 50               | -25.788     | 2.413 | -22.705 | -31.598 |
| Bound-Robin Multi-Age | 0k          | 33               | -20.867     | 0.995 | -18.788 | -22.334 |
| IPPO                  | 0k          | 33               | -21.226     | 1.235 | -18.599 | -23.289 |
| Joint Observation     | 0k          | 33               | -20.268     | 0.806 | -18.567 | -21.394 |
| MAPPO                 | 0k          | 146              | -28.311     | 2.737 | -22.832 | -34.189 |